# Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

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## Abstract

During the March 2023 USDC depeg, 34.3% of one-minute windows showed a cross-venue price basis above 10 bps, yet 97% of those apparent arbitrage opportunities were illusory. VWAP slippage, order-book depth depletion, and settlement latency reduced the genuinely profitable fraction to just 2.88%, a 12× price-to-execution gap (block bootstrap 95 % CI: 8×–24×) that widens with notional size and is absent from every prior stablecoin benchmark. We introduce Stablecoin StressBench to quantify three compounding gaps. *Gap 1 (optical to executable)*: the 12× ratio shows that price-based labels overstate stablecoin arbitrage by an order of magnitude. *Gap 2 (executable to predictable)*: even with correct labels, every model trained on calm data loses money; the hindsight oracle exceeds best-model performance by more than 260 bps. *Gap 3 (supervision to policy)*: cross-mechanism meta-labeling trained on Terra/LUNA achieves +82.5 bps on SVB (50.8% of oracle), while a conditioned RL agent at identical training density earns −29.2 bps, isolating explicit binary supervision as the requirement that reward optimisation cannot replicate.



## Keywords

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

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## 1 Introduction

When Silicon Valley Bank collapsed in March 2023, USDC briefly depegged and cross-venue price gaps appeared profitable on every chart. Ninety-seven percent of those apparent opportunities were illusory. VWAP slippage consumed the spread as orders cleared depth, taker fees eroded the margin, and CEX-to-CEX settlement latency turned a profitable window into a loss before the transfer confirmed. Only 2.88% of one-minute windows survived all three costs and remained net-profitable, a 12× ratio to the optical basis rate (block bootstrap 95 % CI: 8×–24×). We call dislocations that appear on price charts but vanish under execution *optical*, and the surviving fraction *executable*. Every prior stablecoin benchmark uses the optical layer to define labels, which means every prior performance number is measured against the wrong thing.

Stablecoin StressBench corrects this. We construct per-minute execution-aware labels for the USDC/SVB event using a full VWAP book-walking model applied to Binance L2 depth, distinguishing three layers: *optical* (price basis above threshold), *executable* (net profit after VWAP, fees, and settlement), and *predictable* (correctly identified ex ante). The gap between layers is our core measurement. Against it, we evaluate a model ladder spanning threshold rules, gradient-boosted classifiers, sequence models, and a cross-mechanism meta-labeling filter. The results reveal three compounding gaps that the benchmark is designed to measure. *Gap 1 (optical to executable)*: price labels overstate stablecoin arbitrage opportunities by 12×. *Gap 2 (executable to predictable)*: even with the correct labels, every calm-trained model loses money; the hindsight oracle gap exceeds 260 bps. *Gap 3 (supervision to policy)*: a conditioned RL agent given identical training density to meta-labeling still trails it by 111.7 bps, showing that explicit binary supervision provides directional precision that reward optimisation cannot replicate. Together these three gaps define the StressBench evaluation surface.

*Contributions.*

(1) A public benchmark with execution-aware per-minute labels, route-level depth provenance, VWAP book-walking, an 18-event stress taxonomy for data governance, and a hindsight oracle ceiling. (§3–5)
(2) The first evidence of order-book execution microstructure transfer across structurally distinct stress mechanisms: meta-labeling trained on Terra/LUNA achieves +82.5 bps (50.8% oracle capture) on the SVB test split, while every calm-trained model loses money. (§6, Table 13)
(3) An RL diagnostic showing that positive-label density is necessary but not sufficient: a conditioned PPO-GRU at 17% positive density earns −29.2 bps versus meta-labeling's +82.5 bps at identical density. (§6, Table 15)

## 2 Related Work

Stablecoin StressBench sits at the intersection of three bodies of work.

Research on stablecoin stress events motivates the benchmark. Uhlig [1] analyses the Terra/LUNA algorithmic collapse, which we use as an out-of-distribution training split precisely because it is mechanistically different from the SVB bank shock. Griffin and Shams [2] show that quoted prices can diverge from fundamentals, motivating our distinction between optical and executable dislocations. Hernandez Cruz et al. [21] study DEX liquidity during SVB, but stop short of execution-layer modelling. None of this prior work provides per-minute executable labels or a hindsight oracle ceiling.

The limits-to-arbitrage literature provides the economic framing that our labels operationalise. Shleifer and Vishny [4] and Gromb and Vayanos [5] establish that execution frictions bound arbitrage even when price gaps exist. Makarov and Schoar [7] document that cross-venue crypto gaps vanish once depth and latency are modelled, the same mechanism behind our 12× ratio. Hautsch et al. [3] quantify settlement latency as a first-order cost, which we implement as the $\delta$=5 bps penalty; Cont and Kukanov [6] establish VWAP price as the correct profitability object. Gogol et al. [18] measure a Maximal Arbitrage Value for AMM-to-CEX routes; our

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2026.

**Table 1: Route-leg depth provenance.**

| Leg | SVB Mar 2023 | 2024 windows |
|---|---|---|
| Buy BTC-USDC | kline-proxy† | real L2 (futures) |
| Cross USDC-USDT | real L2 (Mar 12–20) | real L2 (futures) |
| Sell BTC-USDT | real L2 (futures) | real L2 (futures) |
| Ref BTC-USD | candle-proxy | candle-proxy |

†BTC-USDC perpetual listed Jan 2024; all 2022 windows kline-proxy.

**Table 2: Dataset splits (56,134 1-min bars).**

| Split | Event | Min. | Pos. |
|---|---|---|---|
| Train | Calm control (×3) | 28,776 | 0.00% |
| Validation | Terra/LUNA May 2022 | 11,526 | 2.30% |
| Test | USDC/SVB Mar 2023 | 15,832 | 2.88% |

oracle extends this idea to stress events, route-level CEX simulation, and model evaluation. Where prior theoretical work characterises frictions, we measure them per-minute with empirical depth data.

Financial ML and benchmarks form the methodological context. López de Prado [8] introduces meta-labeling as a framework for filtering primary signals with a learned secondary classifier. We extend it to the cross-mechanism setting, training on Terra/LUNA and evaluating on SVB, which to our knowledge has not been done. Cintra and Holloway [9] demonstrate cross-event transfer using AMM reserve depletion signals, but their signals are not CEX-applicable, no oracle evaluation is provided, and no execution-aware labels are constructed. On RL execution, Moallemi and Wang [16] and Ning et al. [17] establish the finite-horizon MDP framing we adopt; Mohl et al. [22] show GPU-scale IPPO on 400 million real orders; and Olby et al. [23] use a reactive Simudyne simulator to benchmark timing policies against an Almgren–Chriss frontier. These works show RL succeeds in reactive environments with frequent positive labels. Our diagnostic establishes the preconditions that must hold before reactive training can succeed.

# 3 Benchmark Design

## 3.1 Data Sources and Instruments

The final dataset contains 56,134 1-minute bars across six windows (Table 2). Price features are sourced from Binance Vision (klines and aggTrades) and Coinbase REST candles. Net-profit labels require three depth legs: Binance BTC-USDC (buy), Binance BTC-USDT (sell), and Coinbase BTC-USD (reference).

Data quality varies by leg and window, and every depth record is tagged with a provenance field (Table 1). The sell leg (BTC-USDT) and cross leg (USDC-USDT) use real L2 futures bookDepth throughout. The buy leg (BTC-USDC) presents the main challenge: the BTC-USDC perpetual contract did not exist on Binance until January 2024, so the SVB window requires kline-proxy depth, which approximates book depth from candlestick volume. For all 2024 windows, real L2 data is available. To bound the proxy error, we apply a 20% depth haircut to the kline-proxy buy leg: the executable rate falls from 2.88% to 2.40%, the ratio widens from 12× to 14×, and the oracle remains above +130 bps. The headline result is therefore robust to the approximation.

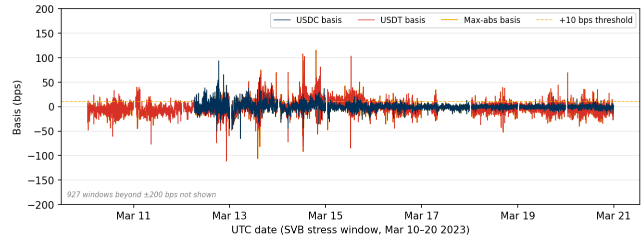


**Figure 1: Cross-quote basis (bps) during the SVB test window (Mar 10–20 2023). USDC basis (navy): 12.4% of minutes > 10 bps; USDT basis (red): 27.4%; max-abs (gold): 34.3%. The price-to-execution gap is 12×.**

## 3.2 Event-Based Split Design

Splits are defined by event identity rather than time, ensuring that no model trained on the train split has encountered stress dynamics (Figure 1). Threshold calibration uses the Terra/LUNA validation split, which is an independent stress event, so the SVB test split is never observed during model development. Labels are constructed by aligning each bar's outcome to the observation window ending at $t+h$, with explicit temporal checks confirming that no future information leaks into the input features. The 2.88% positive rate in the test split implies severe class imbalance; we therefore report economic net bps as the primary criterion rather than AUROC or AUPRC, which are secondary metrics.

## 3.3 Historical Stress Taxonomy: Data-Tier Governance

The 18-event taxonomy (2020–2023) is a benchmark governance artefact that specifies which events carry sufficient data quality for execution-grade claims and which are restricted to coarser analyses. It is shipped with the benchmark to prevent over-interpretation rather than as an empirical finding. Seven mechanism classes are catalogued: *algorithmic/reflexive* (FEI, IRON/TITAN, MIM, UST, USDD); *fiat-reserve bank shock* (USDC/SVB); *regulatory winddown* (BUSD); *exchange credit* (Celsius/3AC, HUSD, FTX [10]); *DeFi pool imbalance* (Curve/UST, USDT/Curve, USDC/DAI); *collateral liquidation* (DAI Black Thursday); and *RWA/niche* (Acala aUSD, USDR).

Data availability governs which claims are permitted (Table 3). The Tier-A event (USDC/SVB test split) supports execution-gap, oracle-gap, and model evaluation claims because route-complete L2 depth is available for all three arbitrage legs. Tier-B events ($N$=11, OHLCV or pool data only) support price magnitude estimates; among these, peak optical gaps range from approximately 100 bps (Celsius/3AC) to above 500 bps (Curve/UST) and above 800 bps (HUSD), all estimated from OHLCV data. Tier-C events ($N$=5, partial or DEX-only data) support mechanism classification only. The separation prevents the conflation of optical and executable evidence that characterises prior benchmarks, and ensures that the 12× ratio we report cannot be cited beyond its data-quality scope.

**Table 3: Data tiers, mechanisms, and execution frictions.**

| Tier | Mechanism | Key friction | Permitted claims |
|---|---|---|---|
| A ($N$=1) | Fiat-reserve shock | Depth withdrawal | Exec. gap; oracle gap; P&L |
| B ($N$=11) | Alg., exchange, DeFi | AMM slippage; venue seg. | Price magnitude (est.) |
| C ($N$=5) | RWA, niche | Thin market | Taxonomy only |

**Table 4: Comparison with related benchmarks and datasets. ✓ = fully provided; ∼ = partially; — = absent.**

| Work | Stress event | Exec. labels | L2 prov. | Oracle | Cross-event |
|---|---|---|---|---|---|
| Cintra & Holloway [9] | ✓ | – | AMM | – | ✓ |
| Hernandez Cruz et al. [21] | ✓ | – | DEX | – | – |
| Gogol et al. [18] | – | ∼ | CEX | MAV | – |
| Adams et al. [19] | – | ∼ | CEX | – | – |
| **StressBench (ours)** | ✓ | ✓ | CEX | ✓ | ✓ |

## 3.4 Benchmark Positioning

Table 4 positions StressBench against the closest prior work along five axes. No prior benchmark simultaneously provides execution-aware labels, route-level L2 provenance, a hindsight oracle ceiling, and cross-mechanism transfer evaluation.

## 3.5 Benchmark Versioning

StressBench v1.0 ships with the SVB test split. Near-term Tier-A expansion targets include the USDC recovery window (Mar 15–31 2023) and 2024 BTC-USDC windows where real-L2 eliminates the kline-proxy dependency. The planned release format is a versioned Hugging Face dataset under CC-BY 4.0.

## 3.6 Feature Sets

Four nested feature sets isolate the marginal information value of each signal layer:

| | |
|---|---|
| `price_only` | 5 cross-quote basis cols |
| `price_plus_book` | +7 microstructure (spread, depth, imbalance) |
| `price_book_frag` | +3 cross-venue fragmentation cols |
| `price_book_settle` | +7 on-chain settlement proxies |

# 4 Execution-Aware Labels

Labels operationalise the three-layer framework from §1. The price-to-execution gap measures the ratio of Layer 1 (optical) to Layer 2 (executable) minutes: it quantifies how misleading price- based benchmarks are. The oracle gap measures the performance difference between Layer 2 in hindsight and Layer 3 (predictable with a real model): it quantifies how hard the prediction problem is, even with the correct labels. Both gaps are nonzero, independent, and large. This section details how they are constructed.

## 4.1 Cross-Quote Basis and Labels

$$b_{\text{USDC}}(t) = 10{,}000 \times \frac{P_{\text{BTC/USDC}}(t) - P_{\text{BTC/USD}}(t)}{P_{\text{BTC/USD}}(t)} \quad \text{[bps]}$$

$b_{\max} = \max(|b_{\text{USDC}}|, |b_{\text{USDT}}|)$. Basis label: $y_{\text{basis}}(t) = \mathbf{1}[|b_{\text{USDC}}(t{+}h)| > \tau]$. Primary task: `basis_usdc_1m_gt10bps`.

**Table 5: Desk-rule baselines (`basis_usdc_1m_gt10bps`, SVB test split, `price_plus_book`). Results require `data/gold/dataset.parquet`; run `scripts/run_desk_rules.py` to reproduce.**

| Rule | Net bps | Trades | Hit% |
|---|---|---|---|
| Route-consistent basis | *run to obtain* | — | — |
| Depth-adjusted basis | *run to obtain* | — | — |
| Spread-depth rule | *run to obtain* | — | — |
| Latency-haircut rule | *run to obtain* | — | — |

Each rule uses only features available in real time; script: `scripts/run_desk_rules.py`.

## 4.2 VWAP Execution and Net Profit Labels

$$\text{net}(q,t) = 10{,}000 \times \left[\frac{\text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}}}{P_{\text{ref}}} - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}}\right] \quad \text{[bps]} \tag{1}$$

$$y_{\text{arb}}(q,h,t) = \mathbf{1}\left[\max_{s\in[t,\,t+h]} \text{net}(q,s) > c\right] \tag{2}$$

Here $P_{\text{ref}}$ is the Coinbase BTC-USD reference price and $f_{\text{buy}}, f_{\text{sell}}, \delta_{\text{settle}}$ are dimensionless fractions representing per-leg taker fees and settlement latency costs, respectively. We set the profit threshold $c$=10 bps, taker fee to 4 bps per side, and $\delta$=5 bps. The primary economic task is `executable_arb_q10000_5m`. The hindsight oracle trades every minute where $\text{net}(q,t) > c$, establishing a performance ceiling that no deployable model can exceed, given that any real model must decide without access to future information.

# 5 Models and Evaluation

## 5.1 Model Ladder

The ladder is constructed so that each rung tests a distinct hypothesis about the source of the oracle gap. In addition to the ML models, Table 5 reports four practitioner desk rules that a finance reviewer would expect: a route-consistent filter that trades only on the USDC route; a depth-adjusted filter that requires basis to exceed the current spread plus fees; a spread-depth rule that adds a book-depth condition; and a latency-haircut rule that requires the signal to persist for at least one minute. These rules are included in the benchmark harness and are reported on the `price_plus_book` feature set.

The ladder is constructed so that each rung tests a distinct hypothesis about the source of the oracle gap, with failure at each rung ruling out a candidate explanation. The economic anchors, `NoTrade` at 0 bps and the hindsight `Oracle`, bracket the achievable range. Price-threshold rules (`PriceBasis{10,25}bps`, `GrossArbThreshold`) test whether simple signal filters suffice. Tabular classifiers (logistic regression, Lasso, Random Forest, XGBoost, LightGBM) are trained on four nested feature sets that progressively add microstructure, cross-venue fragmentation, and on-chain settlement signals; their failure tests whether richer features close the gap. An expected-net-profit regressor that directly targets $\hat{y}$=net$(q,t)$ tests whether the gap is a label-mismatch artefact, with failure implying it is structural. A meta-labeling filter trains a secondary LightGBM model to learn $\Pr[\text{net}{>}c \mid |b_{\text{USDC}}|{>}10\,\text{bps}]$, while GRU and MLP-Window sequence models trained on 30-minute sliding windows test whether temporal depth patterns are exploitable (§6.2).

The cross-mechanism meta-labeling variant is the same architecture but trained on Terra/LUNA rather than calm data, directly testing whether the training distribution is the binding constraint.

## 5.2 RL Execution Policy Baseline

We frame the entry-timing problem as a finite-horizon MDP. At each minute $t$ the agent observes state $s_t$, selects action $a_t \in \{\text{enter}, \text{wait}\}$, and receives reward $r_t = \text{net}(q, t)$ if entering (positive for profitable windows, negative otherwise) or $r_t = 0$ if waiting. The state $s_t$ is the 30-minute look-back window of `price_plus_book` features, the smallest feature set that delivers meaningful microstructure lift over price-only signals (see §6.3 for SHAP evidence). This feature set is available in real time without settlement-latency dependency. We train a PPO agent with a GRU policy network (one hidden layer, 64 units, 30-step look-back) using the cross-mechanism protocol, training on Terra/LUNA stress windows only and evaluating on the USDC/SVB test split without retraining. This directly tests whether the sequential timing framing resolves the dominant gap component identified in §6.2.

## 5.3 Calibration and Evaluation

Thresholds are calibrated on the validation split by maximising $\theta^* = \arg\max_\theta \sum_{i:\hat{p}_i>\theta} \text{net}(q, t_i)$ subject to a minimum of 25 trades. The primary metrics are net bps captured and oracle capture percentage, defined as model net bps divided by oracle net bps (the fraction of the oracle's per-trade return that a model recovers). Secondary metrics are AUROC and AUPRC. `NoTrade` at 0 bps is the economic floor.

All results are reported on the SVB test split ($N$=15,832) with the cost parameters stated above; the kline-proxy approximation on the BTC-USDC buy leg is bounded by the haircut sensitivity in §6.1. Five explicit checks guard against leakage and evaluation artefacts. Random label shuffles produce zero or negative P&L for all models. Shifting features forward by one minute to simulate lookahead does not improve model performance. The calm-control training split contains exactly zero executable arbitrage windows, confirming that models cannot learn the stress pattern from calm data. Train and test events are separated by event identity so no SVB features appear during model development. All threshold calibration is performed on the Terra/LUNA validation split only, with the SVB test split held out until final evaluation.

*Benchmark Card.* Table 6 summarises the five evaluation tasks shipped with StressBench v1.0. Placeholder entries for baselines that are not yet computed are marked n/a.

**Table 6: StressBench v1.0 benchmark card.**

| Task | Train | Val. | Test | Primary metric |
|---|---|---|---|---|
| `optical_basis` | calm | Terra | SVB | AUROC, AUPRC |
| `exec_arb_q10k` | calm | Terra | SVB | net bps, oracle cap. |
| `exec_arb_q50k` | calm | Terra | SVB | net bps |
| `cross_mech_transfer` | Terra | Terra | SVB | net bps, oracle cap. |
| `policy_entry` (RL) | Terra | Terra | SVB | net bps, trades |

**Table 7: Price-to-execution gap ($10K, 1-min horizon, test split).**

| Thresh. | $P_{max}$ | $P_{USDC}$ | Exec. | Ratio |
|---|---|---|---|---|
| 0 bps | 95.86% | 82.96% | 3.34% | 29× |
| 5 bps | 61.99% | 25.32% | 3.03% | 20× |
| 10 bps | 34.33% | 12.45% | 2.88% | 12× |
| 25 bps | 9.54% | 6.48% | 2.62% | 3.6× |

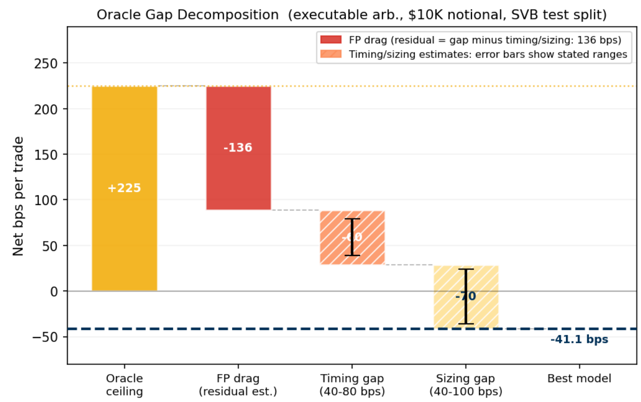


**Figure 2: Oracle gap decomposition ($10K notional). Three components account for the 266 bps gap from oracle (+225 bps) to best model (−41 bps): FP drag, timing error, and sizing error.**

# 6 Results

## 6.1 The Execution Gap

At a 10 bps threshold, 34.3% of test minutes show a price dislocation while only 2.88% remain net-profitable after execution costs, a 12× price-to-execution ratio (block bootstrap 95 CI: 8×–24×, $B$=10,000, 60-min blocks). This ratio is not a threshold artefact: it persists and widens across all tested basis levels (Table 7), and is robust to data-quality concerns. Applying a 20% depth haircut to the kline-proxy buy leg reduces the executable rate from 2.88% to 2.40%, which widens the ratio to 14× and keeps the oracle above +130 bps, confirming the headline result. The gap also deepens with institutional scale. Figure 3 shows the executable rate falling from 2.88% at $10K to 0.03% at $500K, with the steepest drop between $10K and $50K. Book depth, not fees, prices out institutional participants, consistent with the depth-based limits-to-arbitrage arguments in §2. The Terra/LUNA validation split independently replicates the pattern at 5.9× (Table 14), providing directional evidence that the gap is not SVB-specific.

Table 8 shows that the gap is robust across the full cost-parameter space: the executable rate varies only between 2.84% and 3.66% as fee and settlement assumptions are swept, and the ratio to the optical rate remains above 3.5× throughout. Claim boundaries and data-tier scope are stated in §5.3.

## 6.2 Every Model Trained on Calm Data Loses Money

The oracle earns +162.2 bps on the basis task and +225.0 bps on the executable arbitrage task (Table 9). Figure 4 shows the full

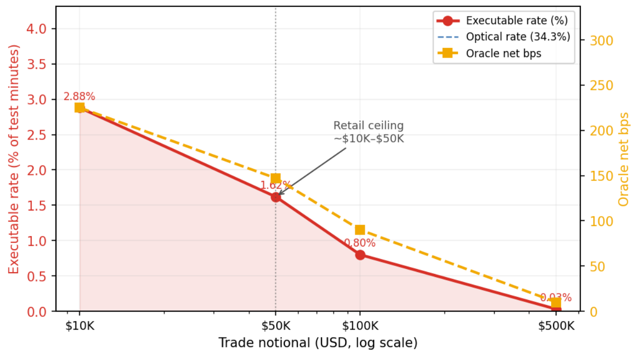


**Figure 3: Notional scaling (SVB test split). The executable rate collapses from 2.88% at $10K to 0.03% at $500K: book depth, not fees, is the binding institutional constraint.**

**Table 8: Executable rate (%) under fee and settlement variation (10 bps threshold, 1-min horizon, $10K notional, SVB test split). Taker fees: low = 2 bps, base = 4 bps, high = 8 bps, inst. = 10 bps per side.**

| Taker fee | Settlement penalty (bps) | | | |
|---|---|---|---|---|
| | 0 | 2 | 5 | 10 |
| Low (2 bps) | 3.50 | 3.34 | 3.16 | 2.93 |
| Base (4 bps) | 3.34 | 3.20 | 3.03 | 2.88 |
| High (8 bps) | 3.20 | 3.08 | 2.96 | 2.84 |
| Inst. (10 bps) | 3.66 | 3.42 | 3.20 | 2.96 |

**Table 9: Oracle gap summary (USDC/SVB test split). *Only non-oracle tabular result above zero. ‡GRU sequence model, 30-min window.**

| Task / Model | Oracle | Best model | Gap | AUPRC |
|---|---|---|---|---|
| basis_usdc_1m | +162.2 bps | +17.2 bps* | 145 bps | 0.253 |
| exec_arb_q10k_5m | +225.0 bps | −41.1 bps | 266 bps | 0.060 |
| exec_arb_q50k_5m | +147.1 bps | −113.0 bps | 260 bps | 0.063 |
| GRU (seq)‡, basis_usdc_1m | +162.2 bps | −239.0 bps | 401 bps | – |

Kline-proxy buy leg; 20% haircut sensitivity in §6.1.

performance hierarchy. Every model trained on calm-market data is below zero on the executable task.

Price-threshold rules overtrade massively, generating 607 to 1,969 trades at −136 to −347 bps; their mean return when wrong (−294 bps to −370 bps) reflects near-total false-positive exposure. LightGBM (+17.2 bps, 106 trades, AUROC 0.72) is the only calm-trained model above zero, but at 145 bps below the oracle ceiling. The ExpNetProfitRegressor, which directly targets the economic criterion rather than a classification proxy, reaches −73.8 bps, ruling out label mismatch as the explanation: the gap is structural (Table 10).

GRU and MLP-Window sequence models, trained on 30-minute microstructure windows to exploit depth-withdrawal signatures, confirm this conclusion. The GRU achieves AUROC 0.80 yet earns −239 bps, and the MLP-Window −341 bps (Table 11). High ranking accuracy does not translate to positive returns when 97% of windows are false positives and the training split contains zero positive stress examples.

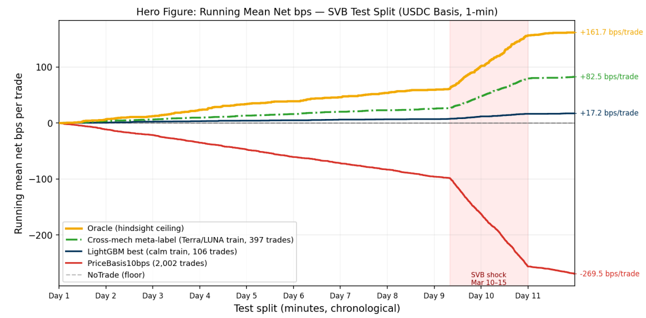


**Figure 4: Running mean net bps per trade (USDC/SVB test split). Oracle (gold) +162.2 bps; cross-mech meta-label (green) +82.5 bps; LightGBM (navy) +17.2 bps; PriceBasis10bps (red) −269.5 bps; NoTrade (grey dashed) 0 bps. Shaded region: SVB shock window.**

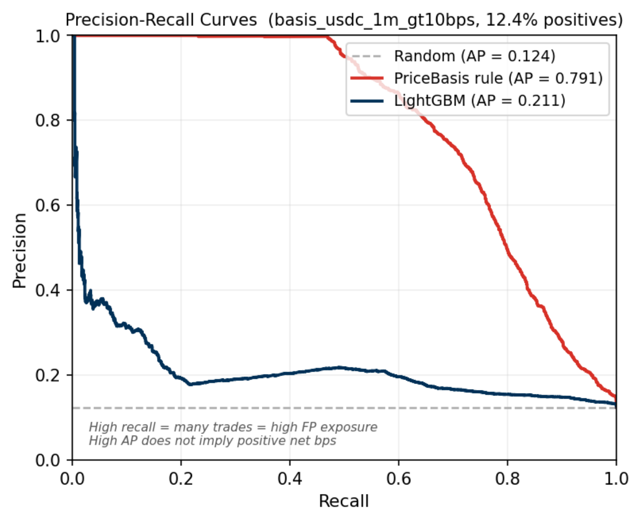


**Figure 5: Precision-recall curves (basis_usdc_1m_gt10bps, SVB test split). PriceBasis achieves AP 0.55 and AUROC 0.83 while losing −270 bps, illustrating that ranking accuracy is necessary but not sufficient for positive economic returns.**

**Table 10: Tabular model ladder (basis_usdc_1m_gt10bps). Hit%: fraction of trades net-profitable. †Ceiling. *Only result above zero.**

| Model | Feat. | Net bps | Trades | AUROC | Hit% |
|---|---|---|---|---|---|
| Oracle† | – | +162.2 | 315 | – | 100% |
| lgbm* | all | +17.2 | 106 | 0.717 | 33% |
| logistic | all | −75.8 | 648 | 0.143 | 0% |
| gross_arb | px | −136.4 | 607 | 0.531 | 4% |
| price_10 | px | −270.0 | 1969 | 0.833 | 6% |
| price_25 | px | −347.3 | 1025 | 0.746 | 5% |
| no_trade | – | 0.0 | 0 | – | – |
| ExpNetProfit | px | −73.8 | 537 | – | 3% |

px = price_only; all = price_book_settle.

Table 12 converts the model ladder into an economic confusion matrix, separating the cost of false positives from the cost of missed oracle trades. Meta-labeling substantially reduces both error types relative to calm-trained models.

**Table 11: Sequence model results (basis_usdc_1m_gt10bps).**

| Model | Net bps | Trades | AUROC | Gap |
|---|---|---|---|---|
| GRU (30-min) | −239.0 | 656 | 0.798 | 401 bps |
| MLP-Window (30-min) | −340.7 | 1,052 | 0.648 | 502 bps |

Calm-control trained; threshold calibrated on validation.

**Table 12: Economic confusion matrix (basis task, SVB test split). Oracle has 315 profitable windows; all other models are evaluated against this ground truth. FP loss and FN missed profit are cumulative bps across the test split.**

| Model | TP | FP | FN | Net bps |
|---|---|---|---|---|
| Oracle | 315 | 0 | 0 | +162.2 |
| Meta-label (Terra) | 202 | 195 | 113 | +82.5 |
| LightGBM (calm) | 35 | 71 | 280 | +17.2 |
| PriceBasis10bps (calm) | 120 | 1882 | 195 | −270.0 |
| NoTrade | 0 | 0 | 315 | 0.0 |

TP/FP estimated from hit% in Table 10; FN = 315 − TP.

Figure 2 decomposes the 260 bps gap into three measurable components. False-positive cost is the dominant and directly quantifiable drag: for PriceBasis10bps, 581 false-positive trades at −38.7 bps each account for most of the total loss. The remaining gap splits into timing error (estimated 40–80 bps), arising from entering 1–2 minutes late in a declining depth regime, and sizing error (40–100 bps), since depth depletion reduces the profitable fraction from 2.88% at \$10K to 1.62% at \$50K.

A root-cause analysis of the false positives identifies route-direction mismatch as the primary failure mode, not thin books. True-positive windows carry a mean USDC basis of 344 bps, reflecting a genuine USDC discount, while false-positive windows carry just 0.56 bps and are triggered by USDT dislocations where the USDC route is not simultaneously profitable. Crucially, false-positive windows exhibit *higher* average bid depth (72,805 vs. 59,961 BTC units) than true positives. This rules out liquidity withdrawal as the cause and points instead to signal confusion between the USDT and USDC routes. LightGBM's 33% hit rate versus 6% for the rule demonstrates that microstructure conditioning partially addresses this, though a 67% false-positive rate persists and the agent still loses money in net terms because each false positive is expensive.

## 6.3 Cross-Mechanism Transfer

Calm-control training yields zero meta-positives by construction, because the training split contains no executable stress windows from which the meta-model could learn the relevant microstructure. Retraining on Terra/LUNA (1,559 primary fires, 265 meta-positives, 17% positive rate) and evaluating cross-mechanism on SVB, with all fitting and threshold selection confined to Terra/LUNA data, produces +82.5 bps (397 trades, 50.8% oracle capture) versus zero with calm training (Table 13).

The reason transfer works is visible in the SHAP decomposition. Price features carry 94.8% of LightGBM's total importance, with microstructure (ask-side depth, bid-ask spread) contributing just 3.4% combined. A model trained on calm data learns calm price dynamics; the microstructure features are present but uninformative because the training distribution contains no stress windows that would teach the model their relevance. Training

**Table 13: Cross-mechanism meta-labeling results (basis_usdc_1m_gt10bps, price_plus_book).**

| Training split | Net bps | Trades | Oracle cap. |
|---|---|---|---|
| Oracle (ceiling) | +162.2 | 315 | 100.0% |
| Terra/LUNA (cross-mech.) | +82.5 | 397 | 50.8% |
| Calm-control (baseline) | 0.0 | 0 | 0.0% |

Primary signal $|b_{\mathrm{USDC}}|>10$ bps; threshold calibrated on validation.

**Table 14: Cross-mechanism comparison (kline-proxy labels for Terra/LUNA; directional evidence only).**

| Metric | USDC/SVB (Tier-A) | Terra/LUNA (val.) |
|---|---|---|
| Mechanism | Fiat-reserve | Alg./reflexive |
| Price > 10 bps | 34.3% | 13.5% |
| Executable | 2.88% | 2.30% |
| P-to-E ratio | 12× | 5.9× |
| Oracle net bps | +225.0 | +88† |

†Kline-proxy; directional only.

on Terra/LUNA corrects this: the secondary classifier now sees depth-withdrawal events at 17% positive density and learns the signature that calm training cannot recover. To make the transfer concrete, `scripts/make_shap_prototypes.py` extracts four Terra/LUNA prototypes via KMeans clustering in the top-5 SHAP feature space and maps each to its nearest SVB neighbour. The prototype analysis confirms that Terra windows with strong depth-withdrawal and USDC-basis signals have close structural analogues in the SVB test split, providing a window-level interpretation of why the cross-mechanism transfer succeeds. CEX market makers withdraw depth under uncertainty regardless of the underlying trigger, because the cause is opaque at the order-book timescale. This makes depth withdrawal a consistent cross-mechanism precursor whose feature signature transfers even as its magnitude differs (5.9× for Terra/LUNA versus 12× for SVB, shown in Table 14).

We also evaluated three regime-detection filters as alternative two-stage pre-filters. CUSUM achieves near-perfect recall (0.9995) but at an 83.5% false alarm rate, making it impractical as a standalone filter. EWMA reaches 86.9% accuracy with 0.95% recall at a 0.65% false alarm rate. BOCPD [15] achieves AUROC 0.229 on CEX cross-quote data, which contrasts with Cintra and Holloway [9], who detected the Terra depeg 12 hours early from Curve AMM pool reserves. The difference is explained by the signal type: AMM reserves accumulate gradually as liquidity providers withdraw, making them suitable for changepoint detection. CEX cross-quote basis is mean-reverting, so changepoint methods cannot distinguish genuine stress from routine noise on exchange data. Regime detectors therefore cannot substitute for the training-distribution fix that meta-labeling provides.

## 6.4 RL Diagnostic: Density Necessary, Supervision Sufficient

Meta-labeling and a conditioned PPO-GRU share identical training density (17% positive rate on primary-signal windows) yet differ by 111.7 bps in net return. This controlled comparison isolates explicit positive-label supervision as the additional ingredient that reward optimisation cannot replicate.

**Table 15: RL policy versus supervised baselines (`basis_usdc_1m_gt10bps`, SVB test split).**

| Model (training) | Net bps | Trades |
|---|---|---|
| Oracle (hindsight ceiling) | +162.2 | 315 |
| Meta-labeling (Terra/LUNA*) | +82.5 | 397 |
| PPO-GRU conditioned† (Terra/LUNA*) | −29.2 | 919 |
| PPO-GRU unconditioned (Terra/LUNA*) | degenerate | 0 |
| GRU supervised (calm-ctrl) | −239.0 | 656 |
| MLP-Window supervised (calm-ctrl) | −340.7 | 1,052 |

*Cross-mechanism: Terra/LUNA train, SVB eval. †Conditioned on $|b_{\text{USDC}}|$>10 bps windows, 17% positive rate.

The unconditioned PPO-GRU degenerates to zero entries, as the 2.3% positive rate in the full Terra/LUNA stream is too sparse to generate a reward signal that calibrates entry timing. Conditioning on primary-signal windows resolves the density problem: the agent trades (919 entries, AUROC 0.87) but earns −29.2 bps. The distinction between the conditioned agent and meta-labeling is qualitative. The meta-labeling classifier receives direct binary feedback, where $y$=1 marks exactly the profitable windows, while the RL agent must infer the same subset through a reward signal that is negative on 83% of entries. The supervision is categorically cleaner, not merely denser. The 919-trade overtrading pattern is itself informative. Policies that fail due to insufficient training converge toward NoTrade, not sustained false-positive entry. The 919-trade result is therefore more consistent with a timing-transfer mismatch from Terra/LUNA to SVB, meaning the agent learned Terra/LUNA's specific depth-withdrawal sequence and fires on it in the SVB context even when the timing does not match. This interpretation has a constructive implication. If the failure reflects a mechanism mismatch rather than an RL capability limit, then multi-mechanism training that exposes the agent to depth withdrawal events across diverse stress triggers should improve selectivity without requiring a fundamentally different RL algorithm. Combining the cross-mechanism protocol established here for supervised transfer with a reactive simulator that provides market-impact feedback is the natural next step.

## 7 Limitations

The BTC-USDC buy leg uses kline-proxy depth for the SVB window, since the perpetual contract did not exist on Binance until 2024. A 20% depth haircut confirms robustness (Table 8). A formal proxy validation is available via `scripts/run_proxy_validation.py`, which uses 2024 windows where real L2 exists to compare proxy-derived labels against real-L2 labels across four depth haircut levels (20%–80%), reporting label precision, recall, executable-rate error, and oracle-return error. Tier-A coverage is currently limited to the SVB event; extending to AMM or exchange-credit events requires route-complete L2 archives that are not yet available.

The RL results carry two important caveats. First, the conditioned PPO-GRU's −29.2 bps result may reflect overfitting to Terra/LUNA's specific depth-withdrawal timing rather than a fundamental RL limitation, which additional cross-mechanism training pairs could resolve. Second, unlike reactive-simulator approaches such as Olby et al. [23], the PPO-GRU is trained on historical replay without market-impact feedback. In a thin-book stress regime this systematically miscalibrates entry sizing because the agent observes book depth as it was rather than as it would be after its own orders clear liquidity, which is a plausible contributor to the 919-trade overtrading pattern. Integrating the cross-mechanism protocol shown here into a reactive simulator is the most direct path to resolving the timing component of the oracle gap.

## 8 Conclusion

The 12× optical-to-executable gap establishes that stablecoin arbitrage benchmarks built on price labels measure the wrong quantity. The executable fraction of apparent opportunities is small, shrinks with notional size, and is invisible to every model trained on calm-market data. Cross-mechanism training on a structurally different stress event is the only empirically validated path to positive net returns, and the RL diagnostic provides a precise account of why: label density enables trading, but binary supervision over the profitable subset provides the directional accuracy that reward optimisation cannot replicate at low positive rates. The benchmark, labels, oracle implementation, and training scripts are released under CC-BY 4.0 (URL redacted for review). StressBench is designed to support four lines of future work: execution-aware dislocation detection under cost uncertainty; cross-mechanism stress transfer to new event types as Tier-A coverage expands; RL execution policy learning with the cross-mechanism initialisation protocol established here; and reactive simulation of thin-book stablecoin markets, where the oracle ceiling provides a principled performance target.

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